

Comparison of Tokenization and Embedding Methods for Disease NER

	Embedding	Tokenizer	Macro F1	Weighted F1
0	Word2Vec	Whitespace	0.4970	0.9307
1	Word2Vec	BPE or WordPiece	0.4720	0.9225
2	Word2Vec	NLTK	0.4880	0.9284
3	GloVe	Whitespace	0.7572	0.9644
4	GloVe	BPE or WordPiece	0.7310	0.9598
5	GloVe	NLTK	0.7440	0.9621
6	FastText	Whitespace	0.5252	0.9343
7	FastText	BPE or WordPiece	0.5080	0.9297
8	FastText	NLTK	0.5190	0.9328
9	ELMo	Whitespace	0.8430	0.9774
10	ELMo	BPE or WordPiece	0.8360	0.9762
11	ELMo	NLTK	0.8390	0.9768
12	BERT	Whitespace	0.8560	0.9794
13	BERT	BPE or WordPiece	0.8580	0.9800
14	BERT	NLTK	0.8424	0.9770

The best-performing model was BERT with BPE/WordPiece tokenization. It achieved the highest Macro F1-score of 0.8580 and Weighted F1-score of 0.9800. BERT also performed strongly with whitespace and NLTK tokenization, but WordPiece gave the best result. This is expected because BERT is designed to work with subword tokenization, which helps handle rare biomedical terms and complex disease names by splitting them into meaningful subword units.

Overall, the results show that contextual embedding methods outperform traditional static embeddings. BERT + WordPiece was selected as the best pipeline because it achieved the highest Macro F1-score. Therefore, it is the most effective approach among the tested methods for disease NER on the BC5CDR dataset.

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Best model based on Macro F1:
Embedding          BERT
Tokenizer          WordPiece
Model              BERT + Softmax
Macro F1           0.858024
Weighted F1        0.980033
Name: 0, dtype: object
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